

Eclogite Facies Reactions, Plagioclase Breakdown & Grossular Garnet

Authoritative References: John D. Winter (Ch 25: Metamorphic Facies) & Dexter Perkins (Ch 13: Garnet Group Mineralogy)

Comprehensive Chapter Masterclass: Eclogite Facies & Garnet Crystal Chemistry

High-Pressure Metabasic Paragenesis, The Breakdown of Anorthite, and the Pyrospite vs. Ugrandite Series

1. Eclogite Facies Petrogenesis & The Destruction of Plagioclase

The **eclogite facies** represents the pinnacle of high-pressure crustal and mantle metamorphism ($P > 1.2\text{--}1.5\text{ GPa}$), equivalent to depths greater than 40–50 km, and temperatures from 500°C to over 900°C). Eclogites are classically defined as bimineralic metabasic rocks consisting essentially of **omphacite** (a jadeite-diopside solid solution sodium-calcium clinopyroxene) and **pyrope-almandine-grossular garnet**, without plagioclase!

"Plagioclase is unstable at the high pressures of the eclogite facies, and its breakdown reactions mark the transition from amphibolite or granulite facies to eclogite facies... Ultimately, anorthite breaks down in the absence of other mineral phases via the reaction: $3\text{ Anorthite} = \text{Grossular} + 2\text{ Kyanite} + \text{Quartz}$."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 25, p. 579)

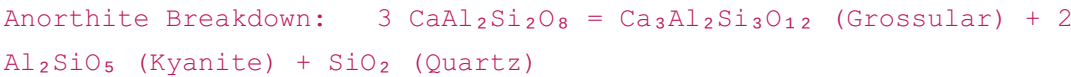
higher-temperature and -pressure eclogites in the western Alps are also mixed in. These are typical eclogite–blueschist associations from subduction zones that are usually tectonically mixed and disrupted following metamorphism. Epidote, zoisite, and quartz, plus hydrous phases such as amphibole, phengite, and paragonite, are common, and the garnets contain <30% pyrope component. Bearth (1959) described an unusual example from near Zermatt, Switzerland, in which deformed (yet recognizable) basaltic pillows exhibit eclogite cores and blueschist rims. Such an occurrence indicates that the protolith was clearly a surface lava and not mantle rock, yet reached very high pressure. The close association of blueschist and eclogite facies rocks has spurred some debate as to whether one mineral assemblage replaces another as prograde or retrograde overprints, or whether they developed under the same (transitional) *T-P* conditions, but with more H₂O available to stabilize the hydrous blueschists. See Yardley (1989) for a discussion.

Group B eclogites occur as lenses in high-grade gneiss terranes. Most are of the medium-temperature type of Carswell and may contain quartz, kyanite, zoisite, paragonite, or Ca-amphibole. The pyrope content of garnet is 30 to 55%. Surrounding rocks are mostly in the amphibolite facies, but there is local evidence for granulite facies and even some blueschist facies rocks. An interesting hypothesis for many of these eclogites is that thrust stacking of crustal units subjected the base of the crust to pressures in excess of 2 GPa and metamorphism in the eclogite facies (see the next section).

Figure 25-12 from John D. Winter (p. 579): ACF compatibility diagram illustrating representative mineral assemblages of the eclogite facies. The shaded region depicts typical basaltic bulk compositions lying in the two-phase field connecting Omphacitic Clinopyroxene (Cpx) and Pyrope-Almandine-Grossular Garnet (Grt), with kyanite appearing in more aluminous varieties.

The Fundamental High-Pressure Breakdown Reactions of Plagioclase

Because plagioclase feldspar contains both albite ($\text{NaAlSi}_3\text{O}_8$) and anorthite ($\text{CaAl}_2\text{Si}_2\text{O}_8$), high lithostatic pressure forces both components to collapse into denser, octahedral-aluminum phases:



Both reactions have positive dP/dT Clapeyron slopes, with the dense product assemblages (lower molar volume) favored at elevated pressures.

2. Mathematical & Stoichiometric Proof of the Reaction

The reaction given in the exam question is: $3 \text{ Anorthite} = \text{Grossular} + 2 \text{ Kyanite} + \text{Quartz}$

Let us perform a rigorous element-by-element atomic inventory to demonstrate that grossular **must** have the stoichiometric formula $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$:

Element	Reactant: 3 × Anorthite ($3 \times \text{CaAl}_2\text{Si}_2\text{O}_8$)	Product: 2 × Kyanite ($2 \times \text{Al}_2\text{SiO}_5$)	Product: 1 × Quartz ($1 \times \text{SiO}_2$)	Residual Required for Grossular	Grossular Atom Count ($\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$)
Calcium (Ca)	$3 \times 1 = 3$	0	0	$3 - 0 = 3$	Ca_3
Aluminum (Al)	$3 \times 2 = 6$	$2 \times 2 = 4$	0	$6 - 4 = 2$	Al_2
Silicon (Si)	$3 \times 2 = 6$	$2 \times 1 = 2$	$1 \times 1 = 1$	$6 - (2+1) = 3$	Si_3
Oxygen (O)	$3 \times 8 = 24$	$2 \times 5 = 10$	$1 \times 2 = 2$	$24 - (10+2) = 12$	O_{12}

The stoichiometric subtraction yields precisely: $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$.

3. Crystal Chemistry of the Garnet Group (Dexter Perkins, Chapter 13)

Garnets are orthosilicates (nesosilicates) belonging to the isometric (cubic) crystal system (space group $\bar{3}d$, point group $4/m\bar{3}2/m$). Their structural formula is written as: $X_3 Y_2 (\text{SiO}_4)_3 \text{ or } X_3 Y_2 \text{Si}_3\text{O}_{12}$ where:

- **\$X\$-site:** Divalent cations in 8-fold (dodecahedral) coordination: Ca^{2+} , Mg^{2+} , Fe^{2+} , Mn^{2+} .
- **\$Y\$-site:** Trivalent cations in 6-fold (octahedral) coordination: Al^{3+} , Fe^{3+} , Cr^{3+} .
- **Tetrahedral site:** Si^{4+} in independent $(\text{SiO}_4)^{4-}$ tetrahedra.

"Mineralogists divide garnets into two series: the pyrospites (pyrope, almandine, spessartine) and the ugrandites (uvarovite, grossular, andradite)... Grossular is the Al end member of the ugrandite series, $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$. Like all garnets, grossular forms equant dodecahedral or trapezohedral crystals and has no cleavage."

— Dexter Perkins, *Mineralogy* (3rd Ed., Chapter 13, pp. 410, 412)

Occurrence and Associations

Spessartine is found with other Mn minerals in Mn-rich skarns, low-grade metamorphic rocks, some rare granites and rhyolites, and, occasionally, in pegmatites. Common associated minerals in igneous rocks are quartz, feldspar, and micas.

Grossular



Origin of Name

From *grossularia*, the Latin name for the pale green gooseberry, which is the same color as some grossular.

Hand Specimen Identification

Like all garnets, grossular forms equant crystals and has no cleavage. Grossular is typically pink, red-pink, or light green. If not, analytical data may be needed to distinguish it from other garnets. See Figure 15.



Description from Dexter Perkins (p. 412): Mineralogical properties, chemical formula ($\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$), and euhedral dodecahedral crystal habit of grossular garnet.



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 9

Question Statement

Q.9: 3 Anorthite = Grossular + 2 Kyanite + Quartz is a reaction which marks the eclogite facies in metabasic rocks. What will be the composition of grossular in the reaction given above?

(a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

(b) CaSiO_3

(c) $(\text{Ca}, \text{Mg})\text{Si}_2\text{O}_6$

(d) CaCO_3

Correct Answer: (a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$



Grossular is the calcium-aluminum end-member of the ugrandite garnet series with the exact stoichiometric formula $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$.

Exhaustive Option-by-Option Critical Analysis

(a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

CORRECT

Why it is uniquely correct: In mineralogy and petrology (Perkins, 2014, p. 412; Winter, 2010, p. 579), **grossular** is universally established as the calcic aluminous end-member garnet with chemical formula $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$. As proven by the mass balance of $3\text{Anorthite} = 3\text{CaAl}_2\text{Si}_2\text{O}_8 \rightarrow \text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12} + 2\text{Al}_2\text{SiO}_5 + \text{SiO}_2$, this formula satisfies stoichiometry exactly.

(b) CaSiO_3

INCORRECT

Why it is incorrect: CaSiO_3 is the chemical formula of **wollastonite**, a calcium inosilicate (pyroxenoid) typical of thermally metamorphosed limestones. Wollastonite lacks aluminum entirely and is structurally and chemically distinct from garnet.

(c) $(\text{Ca, Mg})\text{Si}_2\text{O}_6$

INCORRECT

Why it is incorrect: $(\text{Ca, Mg})\text{Si}_2\text{O}_6$ represents **diopside** (specifically $\text{CaMgSi}_2\text{O}_6$), a monoclinic clinopyroxene. Diopside contains magnesium (absent in pure anorthite) and has a single-chain inosilicate crystal structure with a $(\text{Si}:\text{O})$ ratio of $1:3$, whereas garnet is an orthosilicate with an isolated (SiO_4) ratio of $1:4$.

(d) CaCO_3

INCORRECT

Why it is incorrect: CaCO_3 is the formula for the anhydrous carbonate minerals **calcite** and **aragonite**. Carbonates contain carbon and lack silicon and aluminum altogether.



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Garnet End-Members, High-P Metamorphism & 8 Exam-Grade Multiple-Choice Questions

End-Member Classification & Structural Site Occupancies of the Garnet Group

Garnet Series	Mineral End-Member	Chemical Formula	$\text{X}^{\text{X}}\text{-Site}$ (VIII-fold)	$\text{Y}^{\text{Y}}\text{-Site}$ (VI-fold)	Primary Geological Occurrence
Pyrospite Series (Aluminous)	Pyrope	$\text{Mg}_3\text{Al}_2\text{Si}_3\text{O}_{12}$	Mg^{2+}	Al^{3+}	Mantle peridotites, Kimberlites, Ultra-high-P eclogites
	Almandine	$\text{Fe}_3^{2+}\text{Al}_2\text{Si}_3\text{O}_{12}$	Fe^{2+}	Al^{3+}	Barrovian regional pelitic schists and gneisses
	Spessartine	$\text{Mn}_3\text{Al}_2\text{Si}_3\text{O}_{12}$	Mn^{2+}	Al^{3+}	Granitic pegmatites, Metamorphosed manganese ores
Ugrandite Series (Calcic)	Grossular	$\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$	Ca^{2+}	Al^{3+}	Calc-silicate rocks, Skarns, High-P Eclogites
	Andradite	$\text{Ca}_3\text{Fe}_2^{3+}\text{Si}_3\text{O}_{12}$	Ca^{2+}	Fe^{3+}	Contact metasomatic skarns, Hydrothermal veins
	Uvarovite	$\text{Ca}_3\text{Cr}_2\text{Si}_3\text{O}_{12}$	Ca^{2+}	Cr^{3+}	Chromite-bearing serpentinites, Ultramafic rocks

8 High-Yield Practice Questions on Eclogite Facies & Garnet Mineralogy

1. In the structural formula of garnet $\text{X}_3\text{Y}_2\text{Si}_3\text{O}_{12}$, what are the coordination numbers of the X , Y , and Si sites?

- (a) 4, 6, 8 (b) 8, 6, 4 (c) 6, 6, 6 (d) 12, 8, 4

Answer: (b) 8, 6, 4 — The divalent X^{2+} -site is in 8-fold (dodecahedral) coordination, the trivalent Y^{3+} -site is in 6-fold (octahedral) coordination, and Si^{4+} is in 4-fold (tetrahedral) coordination.

2. Which clinopyroxene is the diagnostic phase defining the eclogite facies in metabasites?

- (a) Diopside (b) Hedenbergite (c) Omphacite (d) Aegirine

Answer: (c) Omphacite — Omphacite is an intermediate solid solution between augite/diopside and jadeite, characteristic of eclogites.

3. The albite breakdown reaction marking the onset of high-pressure blueschist/eclogite conditions is:

- (a) Albite = Nepheline + Quartz
(b) Albite = Jadeite + Quartz
(c) Albite + Calcite = Anorthite + CO_2
(d) Albite = Glaucophane + H_2O

Answer: (b) Albite = Jadeite + Quartz — This classic univariant reaction represents the breakdown of sodic plagioclase at high pressure.

4. The green color of the ugrandite garnet uvarovite is caused by the presence of:

- (a) Titanium (b) Chromium (Cr^{3+}) in the octahedral site (c) Nickel (d) Copper

Answer: (b) Chromium — Uvarovite has the formula $\text{Ca}_3\text{Cr}_2\text{Si}_3\text{O}_{12}$, imparting a vibrant emerald-green color.

5. Coesite and microdiamonds found as inclusions in eclogitic garnets provide proof of:

- (a) Very low temperature (b) Ultra-high pressure (UHP) metamorphism (c) Contact aureole baking
(d) Hydrothermal alteration

Answer: (b) Ultra-high pressure (UHP) metamorphism — Coesite requires $P > 2.5\text{--}3.0\text{ GPa}$ (depths $> 90\text{ km}$), proving subduction of continental/oceanic crust into the mantle.

6. In garnet nomenclature, the name "ugrandite" is a portmanteau derived from:

- (a) Ural, Granite, Diorite
(b) Uvarovite, Grossular, Andradite
(c) Ultramafic, Granulite, Diopside
(d) Underground Anorthite Diopside

Answer: (b) Uvarovite, Grossular, Andradite — The three calcium-bearing end-members form the ugrandite series.

7. In the reaction $3\text{Anorthite} = \text{Grossular} + 2\text{Kyanite} + \text{Quartz}$, why does volume decrease ($-\Delta V$)?

- (a) Garnet and kyanite have much denser, closely packed atomic structures with higher coordination numbers than plagioclase
- (b) Gas is liberated during the reaction
- (c) Quartz contracts under heat
- (d) Aluminum evaporates

Answer: (a) — High pressure favors dense phases where Al enters 6-fold coordination (kyanite and garnet) rather than 4-fold coordination (anorthite).

8. What is the composition of the garnet andradite?

- (a) $\text{Ca}_3\text{Fe}_2^{3+}\text{Si}_3\text{O}_{12}$
- (b) $\text{Fe}_3^{2+}\text{Al}_2\text{Si}_3\text{O}_{12}$
- (c) $\text{Mg}_3\text{Al}_2\text{Si}_3\text{O}_{12}$
- (d) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$

Answer: (a) $\text{Ca}_3\text{Fe}_2^{3+}\text{Si}_3\text{O}_{12}$ — Andradite is the calcium-ferric iron end-member of the ugrandite series.